How to Print a Dual Bellow Stack:

Written by Andrew Heck. Any questions can be sent to andrewlheck@yahoo.com. A more in-depth tutorial on printing a single stack of bellows can be found [here](). The dual stack of bellows is meant for fatigue testing on the cyclic loading test fixture.

1. Create Geometry in SolidWorks

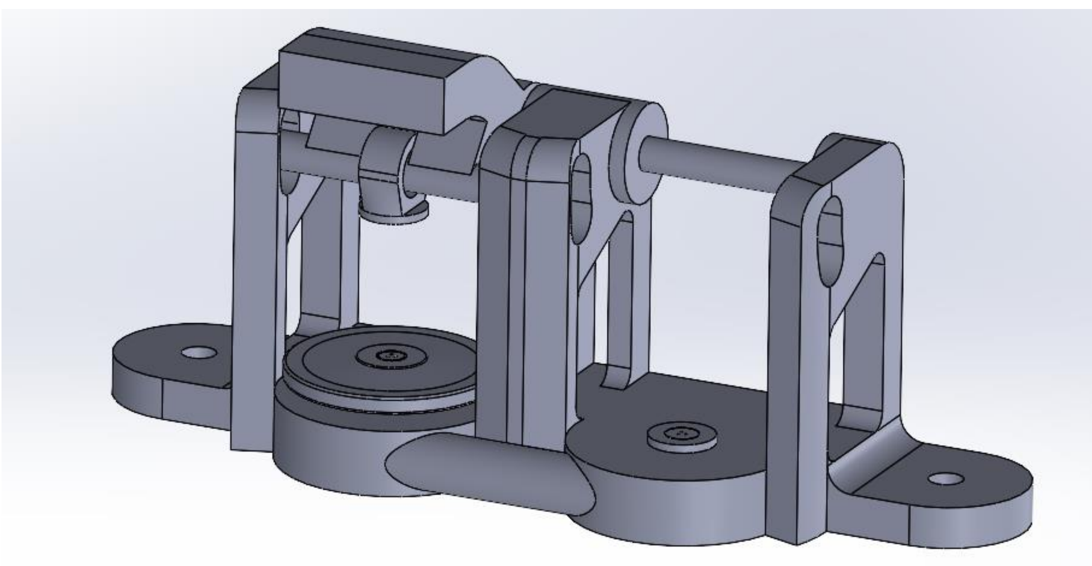
a. Download existing SolidWorks Files

The most up to date CAD models are located in the MACLAB one drive under [MacCurdy Lab/Lab Projects/Printable Hydraulics Designs/SolidWorks Files/](). Download the folder 10.31DoubleStack_Variable_Bellows for a double bellows stack.

b. Generate .STL files

To open the bellows models in OpenVCad you will need to create a .STL file of each individual model in the full bellows assembly. Pre-generated .STL files are in a .zip file called “DualStack_STLs.zip”.

If you want to generate the .STL files yourself, start by opening the top-level bellows assembly (DoubleStack_ASSEMBLY.SLDASM). You should see the model shown below:



While it may look like the model is incomplete it is not. The bellows and legs will be duplicated inside of the OpenVCAD script as it is more efficient to only import one .STL for each repeated component.

To save the model as .STL files go to options: save as, and then set “save as type:” to .stl, make sure that “Output Coordinate System” is set to default, and then select your desired directory. Make sure that **no parts are suppressed** in the assembly when saving .STL files.

2. Generate Model to Print on J750 3D Printer

a. Download OpenVCAD studio

The easiest way to get started with OpenVCAD is with OpenVCAD studio. This is a text editor that will render your models as you run your scripts.

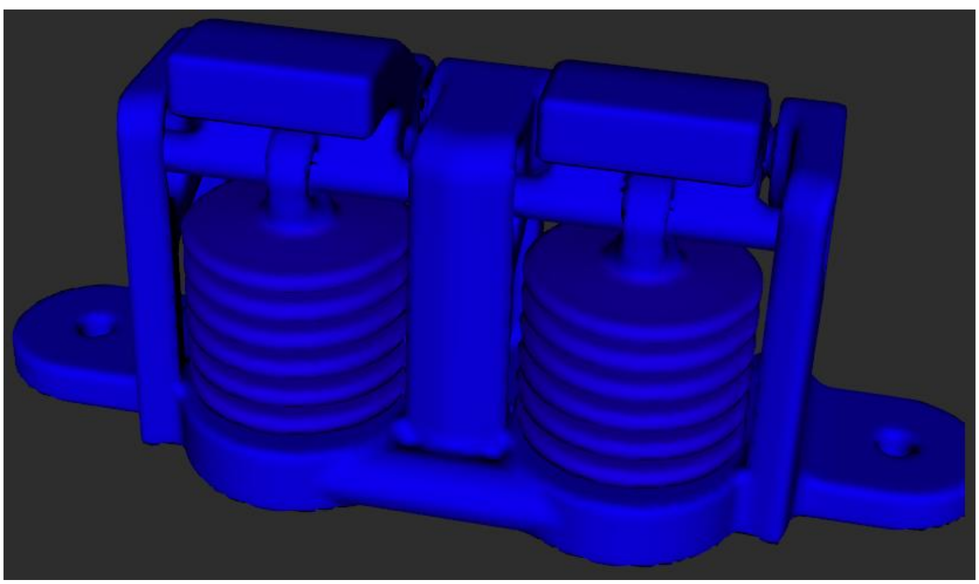
b. Set up your Bellows Script in OpenVCAD Studio

The most up to date script to generate a dual stack of bellows is called “DoubleStackTrial_V2.py” and can be found here.

Open this script directly in OpenVCAD Studio. You will need to update the variable *STL_Location* to where you previously saved your STL files to.

```
#-- STL File Directory --
STL_Location = "saved/to/directory"
```

At this point you should be able to render your dual bellows and see the following:



Feel free to play around with material gradient settings, and use the “Section View” commands at the bottom of the script to look at what the inside of the bellows looks like.